

ISEN/DAEN 427 Lecture 6.2

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Joint Probability Densities of Continuous Random Variables



Motivation

- ▶ In probability theory, we often deal with multiple continuous random variables.
- ▶ Understanding how to combine their probability density functions (PDFs) is essential for modeling joint behavior.
- ▶ Example: measuring height (X) and weight (Y) of a population.

Independent Continuous Random Variables

Definition

Let X and Y be continuous random variables with probability density functions (PDFs) $f_X(x)$ and $f_Y(y)$.

Independence

If X and Y are independent, then their **joint PDF** is the product of the marginals:

$$f_{X,Y}(x,y) = f_X(x) \times f_Y(y)$$

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- ▶ This means the probability structure of X does not depend on Y , and vice versa.
- ▶ The surface $f_{X,Y}(x,y)$ describes how likely each pair (x,y) is to occur.

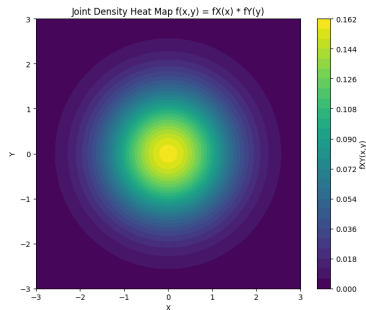
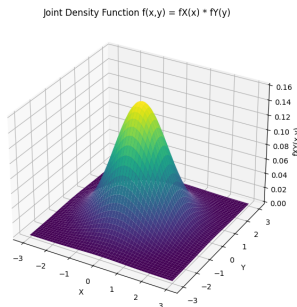
Visualization

- For independent normal distributions:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}}$$

- Their joint PDF is:

$$f_{X,Y}(x,y) = \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}}$$



Dependent Variables

When Variables Are Not Independent

The joint density cannot be written as a simple product. Instead:

$$f_{X,Y}(x,y) = f_{X|Y}(x|y) \times f_Y(y)$$

- ▶ $f_{X|Y}(x|y)$: conditional density of X given $Y = y$
- ▶ Dependence introduces correlation, shaping the joint distribution differently.

Integration to Find Probabilities

Joint Probability Over a Region

For continuous variables, probabilities are areas under the surface:

$$P((X, Y) \in A) = \iint_A f_{X,Y}(x, y) \, dx \, dy$$

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- ▶ The total area under $f_{X,Y}(x, y)$ equals 1.
- ▶ Marginal densities can be obtained by integrating out one variable:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

Summary

- ▶ The joint PDF $f_{X,Y}(x,y)$ describes the full probabilistic relationship between X and Y .
- ▶ For independent variables:

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

- ▶ For dependent variables:

$$f_{X,Y}(x,y) = f_{X|Y}(x|y)f_Y(y)$$

- ▶ Integration allows us to recover marginals or compute probabilities over regions.

Code

```
1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  x = np.linspace(-3, 3, 100)
5  y = np.linspace(-3, 3, 100)
6  X, Y = np.meshgrid(x, y)
7
8  # Define two probability density functions (independent normals)
9  fX = (1/np.sqrt(2*np.pi)) * np.exp(-0.5 * X**2)
10 fY = (1/np.sqrt(2*np.pi)) * np.exp(-0.5 * Y**2)
11
12 fXY = fX * fY # Joint density (multiplication of the two)
13
14 fig = plt.figure(figsize=(8,6))
15 ax = fig.add_subplot(111, projection='3d')
16 ax.plot_surface(X, Y, fXY, cmap='viridis')
17 ax.set_title('Joint Density Function f(x,y) = fX(x) * fY(y)')
18 ax.set_xlabel('X')
19 ax.set_ylabel('Y')
20 ax.set_zlabel('fXY(x,y)')
21 plt.show()
22
23 plt.figure(figsize=(7,6))
24 plt.contourf(X, Y, fXY, levels=30, cmap='viridis') # heatmap
25 plt.colorbar(label='fXY(x,y)')
26 plt.title('Joint Density Heat Map f(x,y) = fX(x) * fY(y)')
27 plt.xlabel('X')
28 plt.ylabel('Y')
29 plt.show()
```